MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **WEEKLY CHAPTER TEST – 02 (LINEAR EQUATIONS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 3: Pair of Linear Equations in Two Variables** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 9(iv).
2. **Section A** comprises 3 Multiple Choice Questions (MCQs) of 1 mark each.
3. **Section B** comprises 1 Assertion-Reason question of 1 mark.
4. **Section C** comprises 2 Very Short Answer (VSAQ) questions of 2 marks each.
5. **Section D** comprises 1 Short Answer (3M) and 1 Long Answer / HOTS (5M) question.
6. **Section E** comprises 1 Applied Case Study with 4 sub-parts of 1 mark each.

SECTION A: Multiple Choice Questions (3 × 1 = 3 Marks)

Q1. If the system of equations $2x + 3y = 7$ and $(a + b)x + (2a - b)y = 21$ has infinitely many solutions, then: [1]

(A) $a = 1, b = 5$ (B) $a = 5, b = 1$ (C) $a = -1, b = 5$ (D) $a = 5, b = -1$

Q2. The value of k for which the system of equations $x + 2y = 3$ and $5x + ky + 7 = 0$ has no solution is: [1]

(A) 10 (B) 6 (C) 3 (D) 1

Q3. If $am \neq bl$, then the system of equations $ax + by = c$ and $lx + my = n$: [1]

(A) has a unique solution (B) has no solution (C) has infinitely many solutions (D) may or may not have a solution

SECTION B: Assertion-Reason Question (1 × 1 = 1 Mark)

Q4. **Assertion (A):** The pair of equations $2x + 3y = 7$ and $kx + \frac{9}{2}y = 12$ has no solution if $k = 3$.
Reason (R): The lines representing $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are parallel if $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION C: Very Short Answer Questions (2 × 2 = 4 Marks)

Q5. Find the value of k for which the pair of linear equations $kx + 3y = -(k - 3)$ and $12x + ky = -k$ has infinitely many solutions. [2]

Q6. Find the values of a and b for which the following system of linear equations has infinitely many solutions: [2]

$$2x + 3y = 7 \quad \text{and} \quad (a + b)x + (2a - b)y = 21$$

SECTION D: Short & Long Answer Questions (1 × 3 + 1 × 5 = 8 Marks)

Q7. Solve the following pair of linear equations graphically: [3]

$$x - y + 1 = 0 \quad \text{and} \quad 3x + 2y - 12 = 0$$

Determine the coordinates of the vertices of the triangle formed by these lines and the x-axis, and shade the triangular region.

Q8. $ABCD$ is a cyclic quadrilateral. Find the measures of all four angles of the cyclic quadrilateral, given that: [5]

$$\angle A = 4y + 20^\circ, \quad \angle B = 3y - 5^\circ, \quad \angle C = -4x^\circ, \quad \text{and} \quad \angle D = -7x + 5^\circ$$

SECTION E: Applied Case Study Based Question (1 × 4 = 4 Marks)

Q9. Case Study: Urban Auto-Rickshaw Fare Structure

In a smart city, the passenger auto-rickshaw fare consists of a fixed hiring charge together with a distance-based charge per kilometre travelled. A student, Aditya, travelled 8 km across City Sector A and paid Rs. 91. On another day, for a continuous commute of 14 km, he paid Rs. 145. Let the fixed charge be Rs. x and the running charge per kilometre be Rs. y.

Based on the above information, answer the following questions:

(i) Formulate the pair of linear equations in x and y representing the two journeys. [1]

(ii) Find the fixed charge (x) and the running charge per kilometre (y). [1]

(iii) If a passenger travels a distance of 25 km, calculate the total fare payable. [1]

(iv) If a passenger paid an amount of Rs. 235, find the total distance travelled by him. [1]

OR

Find the difference between the running charge for 18 km and the fixed charge. [1]

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